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UNIVERSITI MALAYSIA PAHANG
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ASSIGNMENT :
BASIC FIRE SAFETY AND PROPER USE OF FIRE EXTINGUISHERS

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Declaration	We hereby declare that the work is entirely our own effort. Under no circumstances did we allow anyone to copy our work. We understand and accept that any breach of trust will automatically penalize all parties and zero marks will be given for the lab.	

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1. INTRODUCTION TO BASIC FIRE SAFETY

Fire safety is very important in our daily life because it helps protect people, buildings, and property from fire accidents. Fires can start suddenly due to carelessness with electricity, gas stoves, candles, or open flames. Even a small mistake can lead to a dangerous situation. Knowing basic fire safety rules helps us prevent fires and teaches us how to stay calm and act quickly during emergencies. Fire alarms and smoke detectors give early warnings, while fire extinguishers help control small fires before they spread. It is also important to know emergency exits, keep them clear, and practice fire drills regularly. Learning and following fire safety rules is simple, but it plays a big role in saving lives and keeping everyone safe. Everyone should take responsibility for fire safety at home, school, and work. Being prepared can reduce panic and injuries during a fire. Fire safety awareness helps create a safer environment for all.

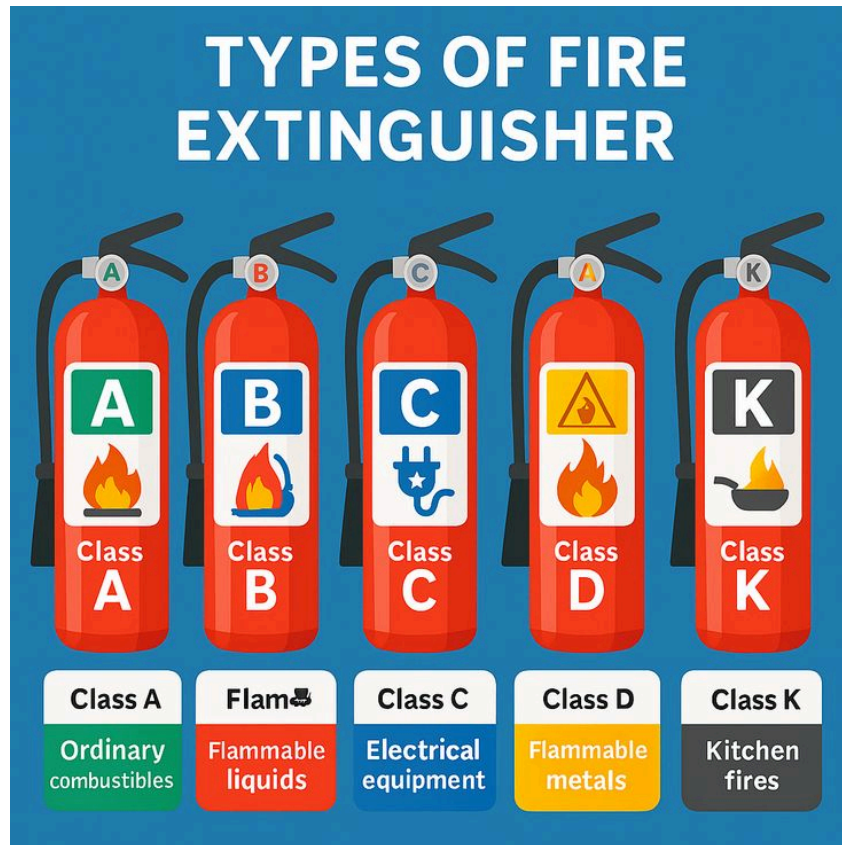
Fire safety also helps reduce damage to property and the environment. Simple habits like switching off appliances can prevent fires. By following fire safety rules, we can protect ourselves and others.

2. EXPLANATION OF FIRE EXTINGUISHERS

Fire extinguishers are important safety tools used to control or put out small fires before they become large and dangerous. They are usually found in homes, schools, offices, factories, and public places. Fire extinguishers work by removing heat, cutting off oxygen, or stopping the chemical reaction of a fire. It is very important to use the correct type of extinguisher for each type of fire to avoid making the situation worse.

Types of Fire Extinguishers:

Types	Description
Class A:	These extinguishers are used for fires involving solid materials such as wood, paper, cloth, rubber, and plastic. Water or foam extinguishers are commonly used for this class.
Class B:	These are used for fires caused by flammable liquids like petrol, diesel, oil, paint, and grease. Foam, dry powder, or CO ₂ extinguishers are suitable for these fires.
Class C:	These extinguishers are used for fires involving flammable gases such as LPG, propane, and butane. Dry powder extinguishers are usually used for this type.
Class D:	These are special extinguishers used for fires involving flammable metals like magnesium, sodium, and potassium. Special dry powder is used for these fires.
Class K:	These extinguishers are used mainly in kitchens for fires involving cooking oils and fats. Wet chemical extinguishers are best for this class.



How to Use a Fire Extinguisher (PASS Method):

- Pull the pin to break the seal and unlock the extinguisher.
- Aim the nozzle at the base of the fire, not at the flames.
Squeeze the handle slowly to release the extinguishing agent.
- Sweep the nozzle from side to side while aiming at the base of the fire.
- Continue sweeping until the fire is completely out.
- Using fire extinguishers correctly can help save lives and reduce damage during emergencies.

HOW TO USE A FIRE EXTINGUISHER

P A S S



Pull the pin
in the handle



Aim the nozzle
at the base
of the fire

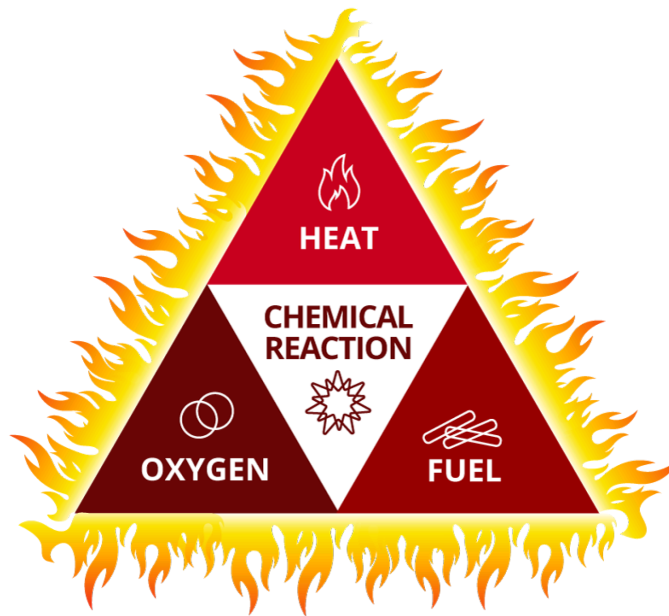


Squeeze the
lever slowly



Sweep from
side to side

3. FIRE TRIANGLE AND FIRE TETRAHEDRON



FIRE TRIANGLE

The fire triangle explains the three basic elements needed for a fire to start and continue burning. These three elements are heat, fuel, and oxygen. Heat provides the energy needed to start a fire. Fuel is any material that can burn, such as wood, paper, gas, or oil. Oxygen supports the burning process and usually comes from the air. If any one of these three elements is removed, the fire will stop.

FIRE TETRAHEDRON

The fire tetrahedron is an extended form of the fire triangle. It includes a fourth element called the chemical chain reaction. This reaction helps the fire to continue burning once it has started. The four elements of the fire tetrahedron are heat, fuel, oxygen, and chemical chain reaction. Fire can be controlled or extinguished by removing any one of these four elements. Understanding the fire triangle and fire tetrahedron helps us know how fires start and how they can be safely put out.

4. FIRE EXTINGUISHER DEMONSTRATION

1. Stay Calm and Assess the Situation

- Before using a fire extinguisher, remain calm.
- Check if the fire is small and manageable.
- Ensure you have a clear exit path behind you in case the fire spreads.

2. Stand at a Safe Distance

- Before using a fire extinguisher, remain calm.
- Check if the fire is small and manageable.
- Ensure you have a clear exit path behind you in case the fire spreads.

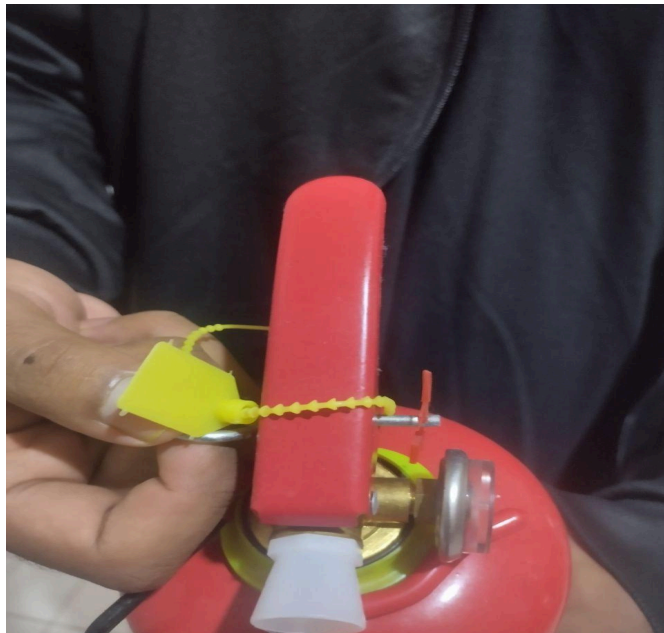


3. Identify the Correct Fire Extinguisher

- Make sure you are using the right type of extinguisher for the fire class (A, B, C, D, or K).
- Using the wrong extinguisher can be dangerous and may spread the fire.

4. Pull the Safety Pin

- Hold the extinguisher firmly and pull the pin to break the safety seal.
- This unlocks the handle so you can operate the extinguisher.



5. Hold the Extinguisher Upright and Point the Nozzle

- Keep the extinguisher upright with both hands.
- Point the nozzle or hose toward the base of the fire.

6. Squeeze the Handle Slowly

- Apply even pressure to the handle to release the extinguishing agent.
- Avoid squeezing too fast; a steady flow works best to control the fire.

7. Sweep the Nozzle from Side to Side

- Move the nozzle slowly from left to right across the base of the fire.
- Cover the entire area of the fire until it is completely extinguished.



8. Ensure the Fire is Out

- After the flames disappear, continue to watch the area.
- Some fires can reignite if hidden embers remain.

5. CONCLUSION

Fire safety is a very important part of protecting lives, property, and the environment. Fires can start unexpectedly due to carelessness, electrical faults, flammable materials, or cooking accidents, and even a small fire can quickly become dangerous. By understanding fire safety, we can prevent accidents and respond effectively in emergencies. Learning about fire extinguishers, their types, and proper usage through the PASS method allows us to control small fires safely before they spread.

The fire triangle and fire tetrahedron help us understand how fires start and continue to burn, showing that removing any essential element of heat, fuel, oxygen, or the chemical chain reaction can stop a fire. Practicing fire drills, knowing emergency exits, and maintaining fire safety equipment like alarms and extinguishers create a safer environment at home, school, and workplace.

Proper fire safety knowledge not only saves lives but also reduces property damage and prevents injuries. Everyone, including children, adults, and employees, should take responsibility for fire safety and remain vigilant. Regular training and awareness help people act quickly and calmly during a fire, ensuring safety for themselves and others. In conclusion, fire safety is simple to learn but has a huge impact, and following fire safety rules can prevent disasters and protect lives every day.